rate_control

Problem 3:

Quadcopter Model

Below we answer the questions to problem 3. The implementations are provided in the following python programs:

Pitch axis

File: `pitch_compensator.py`

The script models one rotational axis as a SISO loop after allocating motor command changes through a pitcher mixer.

Roll axis

File: `roll_compensator.py`

Each script models one axis body-expressed rotational rate as a SISO loop with motor commands allocation through a roll mixer.

1- Expected hover RPM

The thrust per each motor is :

$$T_i = C_T \cdot \rho \cdot n^2 \cdot D^4$$

Simplified to

$$T_i \approx k_{thrust} \cdot \omega^2$$

where ω is the angular speed in rad/s

A hover state is when the **net** total forces and moments are zeros

$$\sum_i F_i = 0$$

$$\sum_i M_i = 0$$

The thrust per motor required to balance the weight is:

$$T_{motor} = \frac{m.g}{4}$$

For a total mass of $1.14\ Kg$, the thrust per motor required is $2.795\ N$

From the figure 3-1, the RPM corresponding to required thrust per motor is $14250\ \mathrm{RPM}$

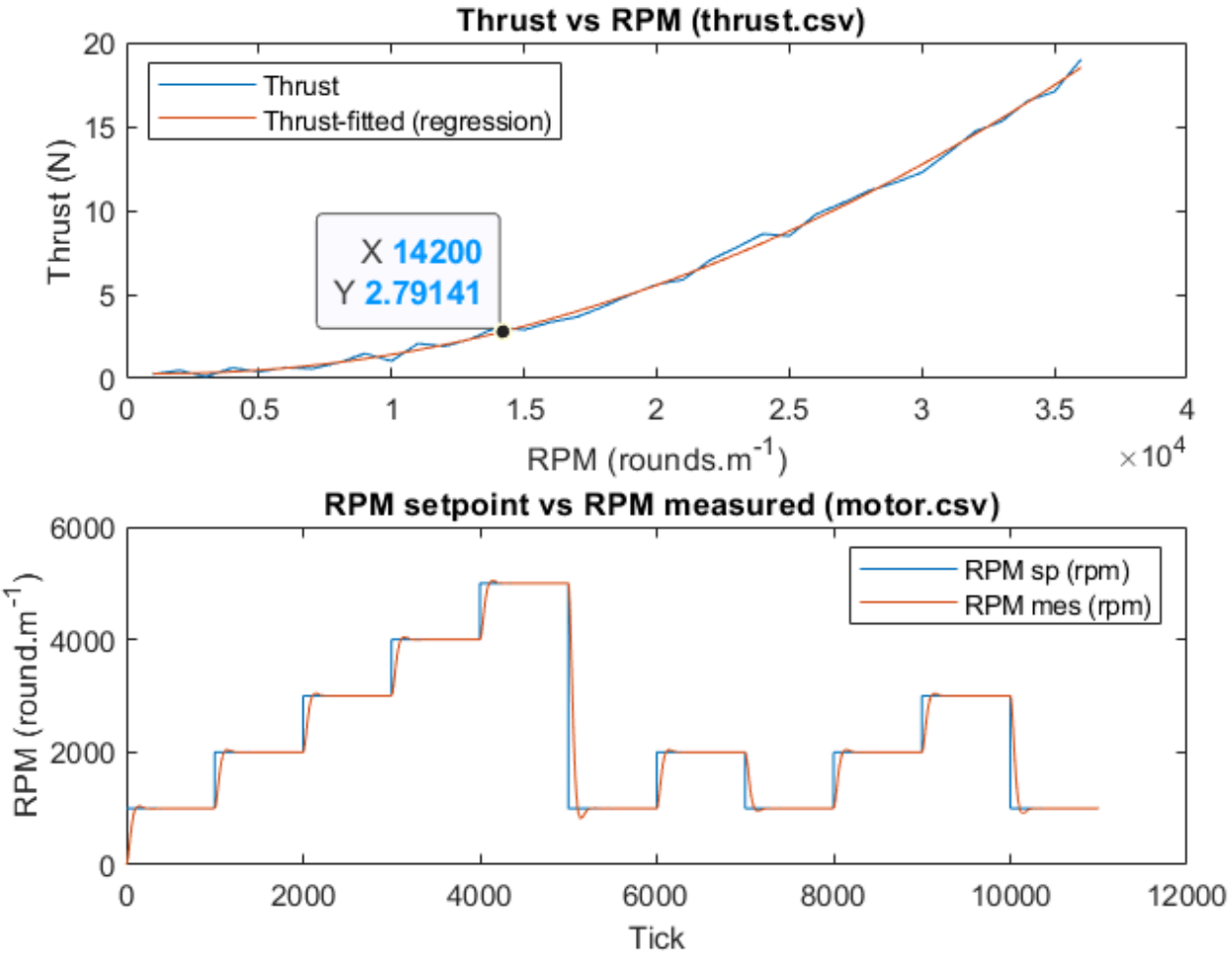


Figure 1-1: Thrust (N) vs RPM

2- Nonlinear MISO models

The drone X configuration is symmetric, so body x-axis is forward across motor 1 & 2 with the origin at the CG, the body y-axis is outward across motor 2 and 3, while z-axis is down through the CG completing the right-hand rule.

Motor 1 (Front-Left): $+x, -y$ [CW]

Motor 2 (Front-Right): $+x, +y$ [CCW]

Motor 3 (Rear-Right): $-x, +y$ [CW]

Motor 4 (Rear-Left): $-x, -y$ [CCW]

The pitch motion occurs when there is a differential between the thrust of $(F_1 + F_2)$ and $(F_3 + F_4)$
The roll motion occurs when there is a differential between the thrust of $(F_1 + F_4)$ and $(F_2 + F_3)$

We assume proportional relationship between the motor thrust and angular speed of the motor :

$$T_i = k_{thrust} \cdot \omega^2$$

where k_{thrust} is the thrust coefficient.

Provided that :

$$L_x = 0.07\,\mathrm{m}$$
$$L_y = 0.09\,\mathrm{m}$$

where L_x and L_y is the moment arm length of the fuselage/frame in body's x and y axes direction, respectively

$$L_{body} = 0.08\,\mathrm{m}$$
$$W_{body} = 0.04\,\mathrm{m}$$

Where L and W are half of the length (assumed) and width of fuselage/frame, respectively.

Assuming a rectangular cuboid frame:

$$J_{xx}(body) = \frac{1}{12} \times m_b W_{body}^2 = \mathbf{1.33 \times 10^{-4} kg.m^2}$$

$$J_{yy}(body) = \frac{1}{12} \times m_b L_{body}^2 = \mathbf{5.33 \times 10^{-4} kg.m^2}$$

$$J_{xx}(motor) = 4 \times m_m L_y^2 = 4 \times 0.035 \times (0.09m)^2 = \mathbf{1.134 \times 10^{-3} kg.m^2}$$

$$J_{yy}(motor) = 4 \times m_m L_x^2 = 4 \times 0.035 \times (0.07m)^2 = \mathbf{6.860 \times 10^{-4} kg.m^2}$$

$$J_{xx} = J_{xx}(body) + J_{xx}(motor) = \mathbf{1.2673 \times 10^{-3} kg.m^2}$$

$$J_{yy} = J_{yy}(body) + J_{yy}(motor) = \mathbf{1.2193 \times 10^{-3} kg.m^2}$$

Establishing the properties associated with mass and moment of inertia. Now we shift to deriving the transfer function of the plants

Recall

$$au = J\dot{\omega} + \omega \times J\omega \approx J\dot{\omega}$$

where τ is the net external torque

Note, tha the gyroscopic effects from the body rotated frame with respect to inertial frame are generally negligible in time scale of controller transient responses

where J is the moment of inertia matrix, since we have assumed previously a symmetry, we have

$$J = \begin{bmatrix} J_{xx} & 0 & 0 \\ 0 & J_{yy} & 0 \\ 0 & 0 & J_{zz} \end{bmatrix} = \begin{bmatrix} J_{xx} & 0 & 0 \\ 0 & J_{yy} & 0 \\ 0 & 0 & J_{xx} + J_{yy} \end{bmatrix}$$

or

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} J_{xx} & 0 & 0 \\ 0 & J_{yy} & 0 \\ 0 & 0 & J_{zz} \end{bmatrix} \begin{bmatrix} \tau_{roll} \\ \tau_{pitch} \\ \tau_{yaw} \end{bmatrix}$$

The pitch rate q:

For pitch moment :

$$\dot{q} = \frac{L_x}{J_{yy}} \Sigma F_i$$

$$\dot{q} = \frac{L_x \cdot k_{thrust}}{J_{yy}} (\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2)$$

The roll p :

For the roll moment

$$\dot{p} = \frac{L_x}{J_{xx}} \Sigma F_i$$

$$\dot{p} = \frac{L_y \cdot k_{thrust}}{J_{xx}} (\omega_1^2 - \omega_2^2 - \omega_3^2 + \omega_4^2)$$

Note that ω_i is in $\mathrm{rad/s}$ and cane be easily expressed in $\mathbf{RPM}$

$$\omega \left[\mathrm{rad/s}\right] = \mathrm{RPM} \, \frac{2\pi}{60}$$

3-Linearization of the MISO models

The motor thrust which creates a lifting force during hover, is a nonlinear function of its velocity as mentioned before

$$F(\omega) = k_{\mathrm{thrust}} \omega^2$$

Aroud hover point, the linearized function is obtained from the first-order tylor series (truncating pther terms):

$$F(\omega) \approx F(\omega_0) + \left. \frac{\partial F}{\partial \omega} \right|_{\omega_0} (\omega - \omega_0)$$

Where ω_0 (rad/s) is the hover angular speed of the motor.

Calculating the derivative with respect to ω:

$$\frac{\partial F}{\partial \omega} = 2k_{thrust}\omega \implies \left.\frac{\partial F}{\partial \omega}\right|_{\omega_0} = 2k_{thrust}\omega_0$$

So:

$$F_i \approx k_{thrust}\omega_0^2 + 2k_{thrust}\omega_0 u_i$$

Note that the control input u_i is the motor's angular speed deviation from the hover angular speed

$$u_i = \Delta\omega_i = \omega_i - \omega_0$$

Substituting in the above pitch ODE equation and eliminating the repeated steady-state terms of $k_{thrust}\omega_0^2$

The input linearized pitch equation becomes:

$$\dot{q} = \left[\frac{2L_x k_{thrust}\omega_0}{J_{yy}}\right](u_1 + u_2 - u_3 - u_4)$$

Similarly, the linearized roll equation becomes:

$$\dot{p} = \left[\frac{2L_y k_{thrust}\omega_0}{J_{xx}}\right](u_1 - u_2 - u_3 + u_4)$$

4- Decoupling MISO models

To make multiple inputs to motors become a single input, we place a mixer stage to allocate and map a single input to the different motor inputs via an mixer matrix.

Actuator Allocation:

Mixed Pitch-rate Input δ_q:

For positive pitch angle : commands the front motors to throttle up, rear motors to throttle down.

$$\delta_q = u_1 + u_2 - u_3 - u_4$$

Mixed Roll-rate Input δ_p:

For positive roll angle: commands the left motors to throttle up, right motors to throttle down.

$$\delta_p = u_1 - u_2 - u_3 + u_4$$

Pitch-rate SISO system

Lumping the cosntant values into η_{pitch} and η_{roll} and taking the laplace transform directly, we obtain

$$\frac{Q(s)}{U_q(s)} = \frac{\eta_q}{s}$$

Roll-rate SISO system:

$$\frac{P(s)}{U_p(s)} = \frac{\eta_p}{s}$$

The system plant is not only the vehicle dynamics but also the motor dynamic response, the lumped motor dynamic response can be reduced to a first-order transfer function form

$$G_{motor} = \frac{1}{T_{motor}s + 1}$$

Where T_{motor} is the motor time constant, taken to be T_{motor} is $50\ \mathrm{ms}$ assuming the sampling time dt

Finally the plant SISO transfer functions are:

$$G_q = \frac{Q(s)}{U_q(s)} G_{motor} = \frac{\eta_q}{s(T_{motor}s + 1)}$$

$$G_p = \frac{P(s)}{U_p(s)} G_{motor} = \frac{\eta_p}{s(T_{motor}s + 1)}$$

where

$$\eta_q = \frac{2 \cdot L_x \cdot k_{thrust} \cdot \omega_0}{J_{yy}}$$

and

$$\eta_p = \frac{2 \cdot L_y \cdot k_{thrust} \cdot \omega_0}{J_{xx}}$$

The mixer matrix:

This is the post controller stage entering the actuation (motors) which maps the single input per angular rate axis to the different motor inputs

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ -1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \delta_p \\ \delta_q \end{bmatrix}$$

5- Poles and control margins of the Transfer Functions

The uncompensated open-loop pitch-rate $G_q(s)$ and $G_p(s)$ are second order type-1 systems.

- $G_q(s)$ **Pitch-Rate Transfer Function**

$$G_q(s) = \frac{0.215}{0.05s^2 + s}$$

There are two poles : $(-20.0, 0.j)$ and at the origin $(0.0 + 0.0j)$

Gain Margin (GM): ∞ (asymptotic approach to -180°)

Phase Margin (PM): 89.38° at $0.22\ \mathrm{rad/s}$

- $G_p(s)$ **Roll-Rate Transfer Function**

$$G_p(s) = \frac{0.266}{0.05s^2 + s}$$

There are two poles : $(-20.0, 0.j)$ and at the origin $(0.0 + 0.0j)$

Gain Margin (GM): ∞ (asymptotic approach to -180°)

Phase Margin (PM): 89.24° at $0.27\ \mathrm{rad/s}$

While the gain margin theoretically is infinity in the continuous-time case implying that the sytem remains stable in terms of gain as it never crosses the $-180\ \mathrm{deg}$ line, once we discretize the system, the potential for infinite gain disappears and becomes a finite number.

Even with such high PM (See figure 1-2 and 1-3), we see that the open-loop gain cross-over frequnecies are very low : at $0.22\ \mathrm{rad/s}$ and $0.27\ \mathrm{rad/s}$ for the $G_q(s)$ and $G_p(s)$ respectively. This suggests that the system is sluggish when commanded, with slow step tracking which is unacceptable for inner-most response, even more alarming if the motor command is sustained it will lead to unbounded accumulated errors.

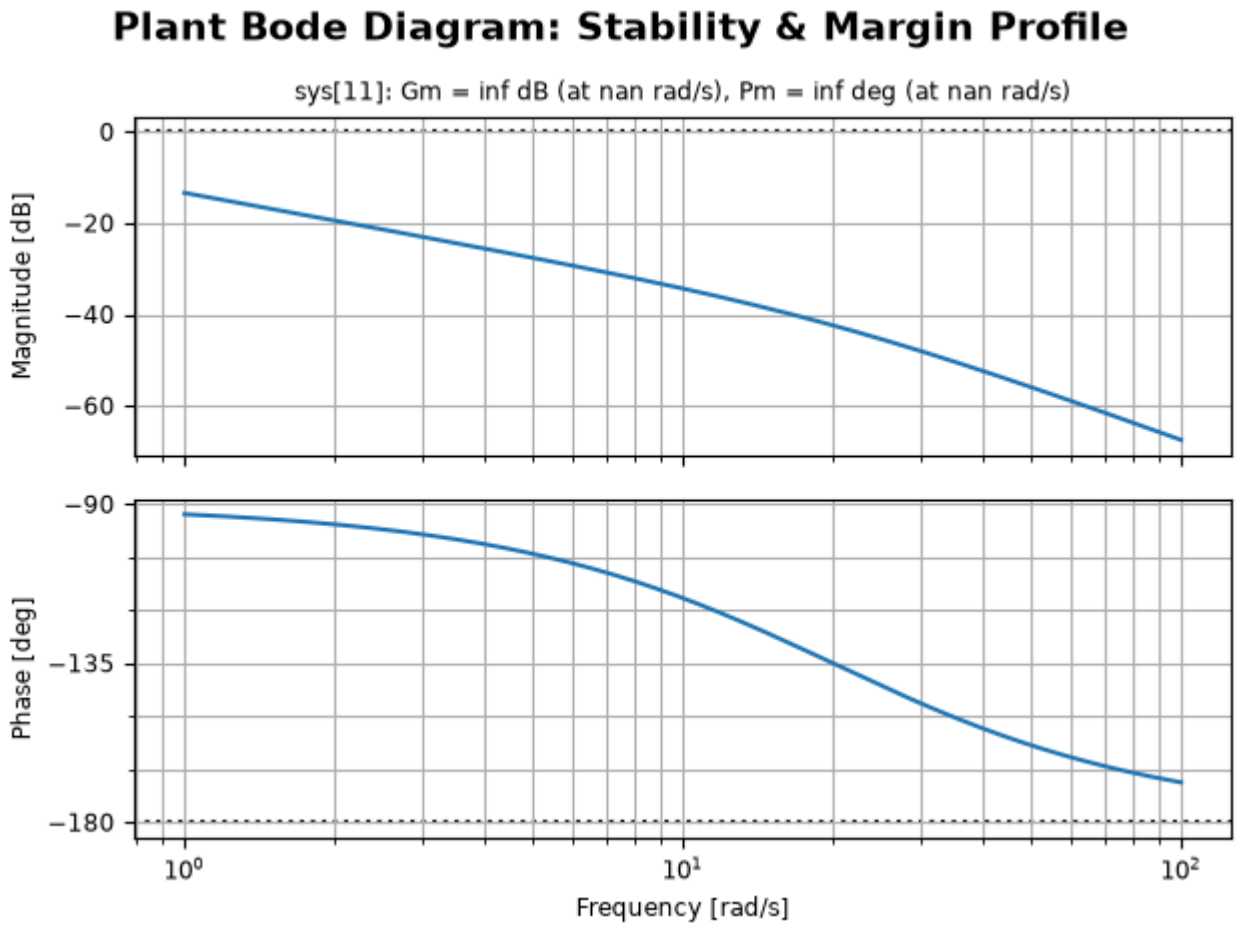


Figure 1-2: Bode plot of the pitch rate plant

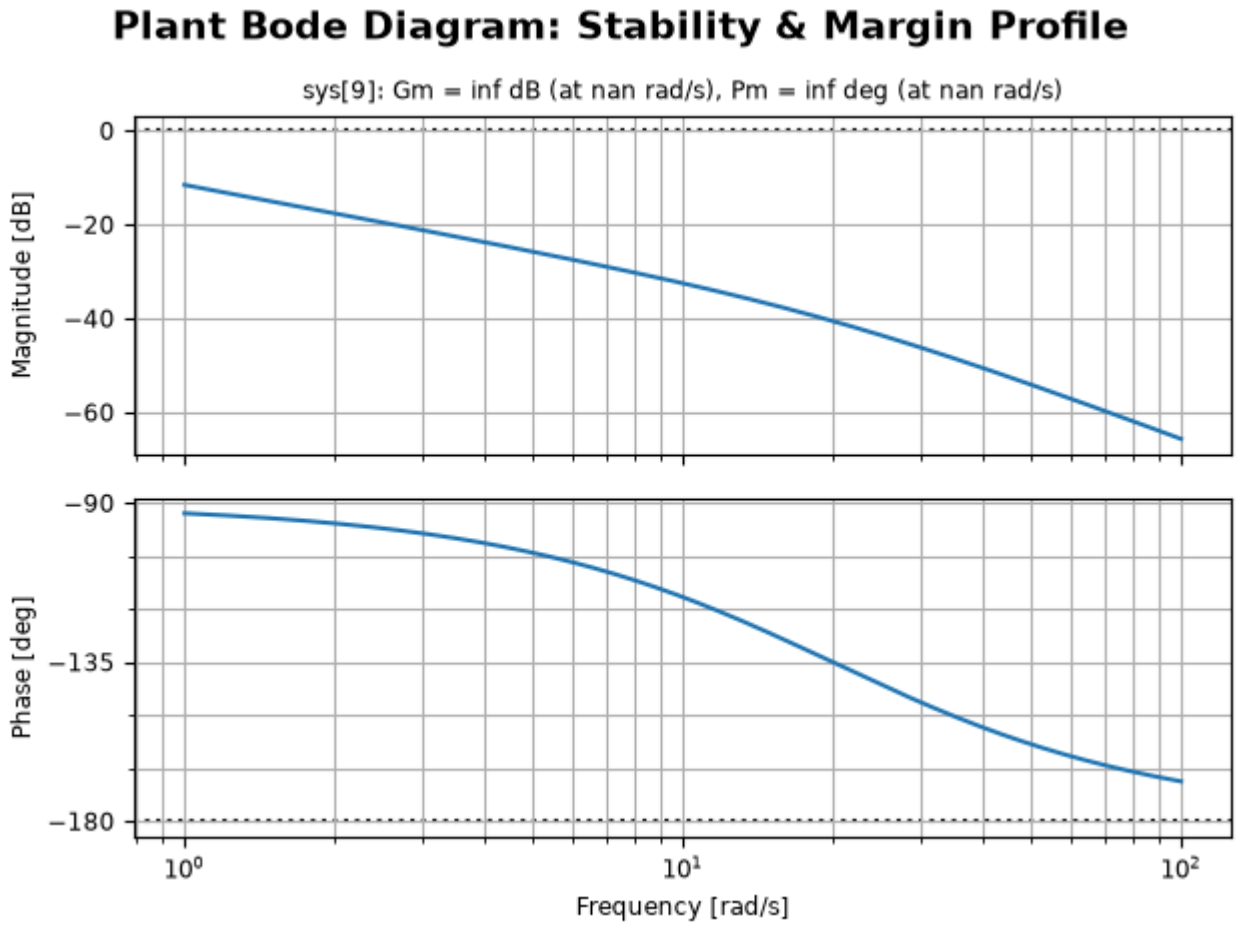


Figure 1-3: Bode plot of the roll rate plant

6- Design a regulator for both rate transfer functions

We can infer some requirements from the "motor.csv" eventhough no time stamps are provided for the RMP setpoints and responses signals. We assume a sampling frequency of $1kHz$.

The rise time and settling time are 80 and 220 ticks respectively, so assuming a sample time of $dt = 1ms$:

Requirement	Value
t_{rise}	$80\,\text{ms}$
$t_{settling}$	$220\,\text{ms}$
$Overshoot$	$< 5\%$
t_{delay}	$1\,\text{ms}$
PM	$> 60^\circ$
GM	$> 6dB$

Control transfer function was designed using phase Lead compensator, which is suitable for damping the type-1 (one integrator term) second order system. Discretization of the lead compensator and transforming it to a PID structure would add an integrator action (See pitch_compensator.py and roll_compensator.py for implementation of the continuous-domain lead compensator).

The structure of the phase lead compensator is :

$$C(s) = K_c \frac{s+z}{s+p}$$

The open-loop transfer function

$$L(s) = C(s)G(s)$$

Transfer function	Pitch-Rate	Roll-rate
$C(s)$	$\frac{621.376(s+17.585)}{s+67.197}$	$\frac{767.539(s+19.100)}{s+89.110}$
$L(s)$	$\frac{2350s+4.134\times10^4}{0.8797s^3+76.68s^2+1182s}$	$\frac{3900s+7.448\times10^4}{0.9549s^3+104.2s^2+1702s}$

Note that the controllers' continuous-domain transfer functions can easily be dsicretized (tustin method) then transformed into an equivalent difference equations relating the control input $u[k]$ to error $e[k] =$

$\text{setpoint} - y[k]$ from which the parallel-form PID terms are obtained.

Frequency Responses:

The primary frequency response change afforded by the lead compensator is a significant upward shift of the entire magnitude curve. The magnitude at 1 rad/s has been raised from roughly $-10dB$ to more $+25dB$ for both transfer functions. This implies the Pitch-rate compensator introduced a high proportional gain K_p to improve tracking and eliminate steady-state error. With this added gain, the gain crossover frequency is at ω_c of $34.38\ \text{rad/s}$ (marked by the vertical dotted line). At this frequency a Phase Margin of 65.98° is observed and it's compliant with the design requirement. On the other hand the Gain Margin G_m remains infinte since thephase asymptotically approaches but never rosses -180°.

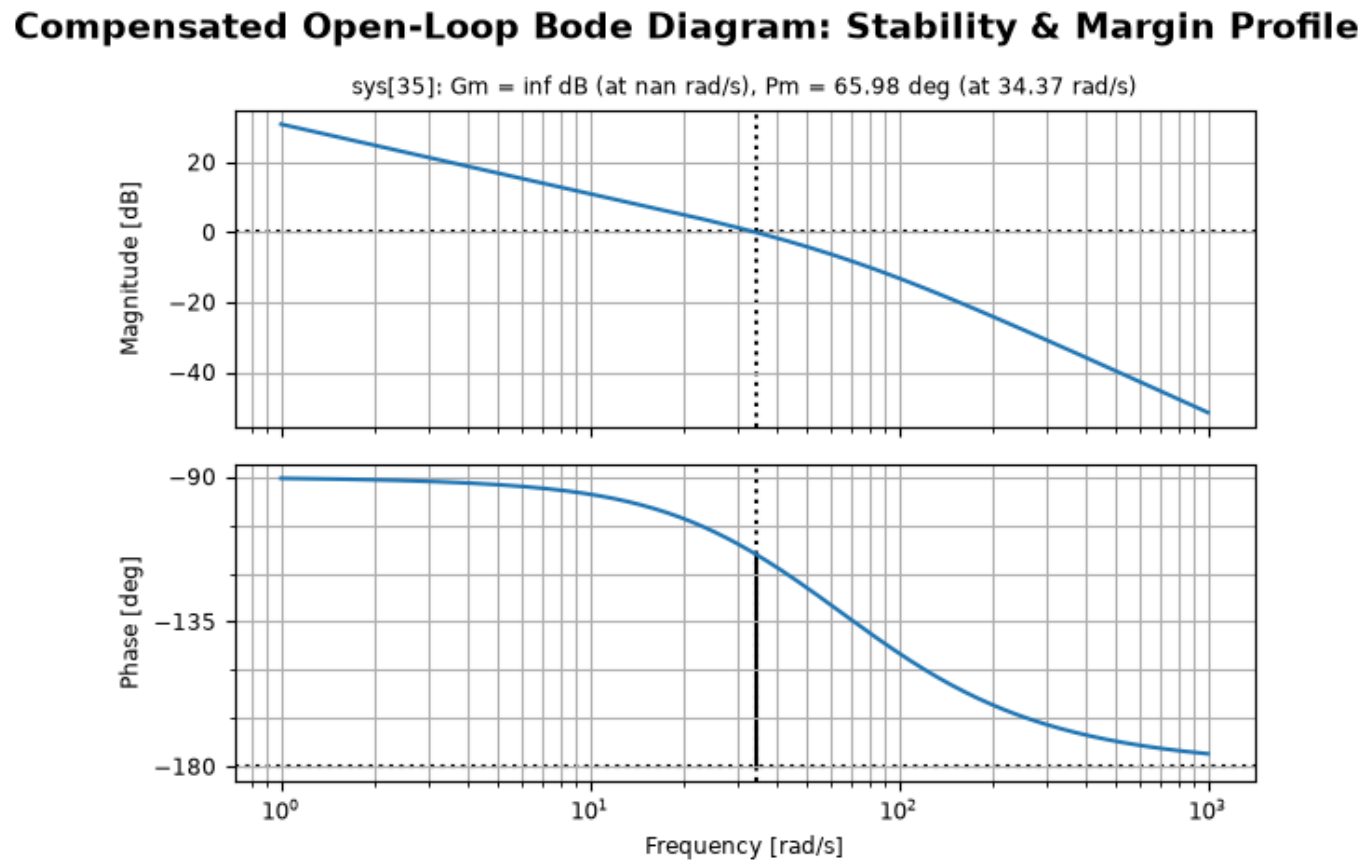


Figure 1-4: Bode plot of compensated Open-loop for pitch rate

In the roll-rate compensator the Gain Margin remained at ∞ while the Phase Margin is 66.18° at $41.25\ \text{rad/s}$.

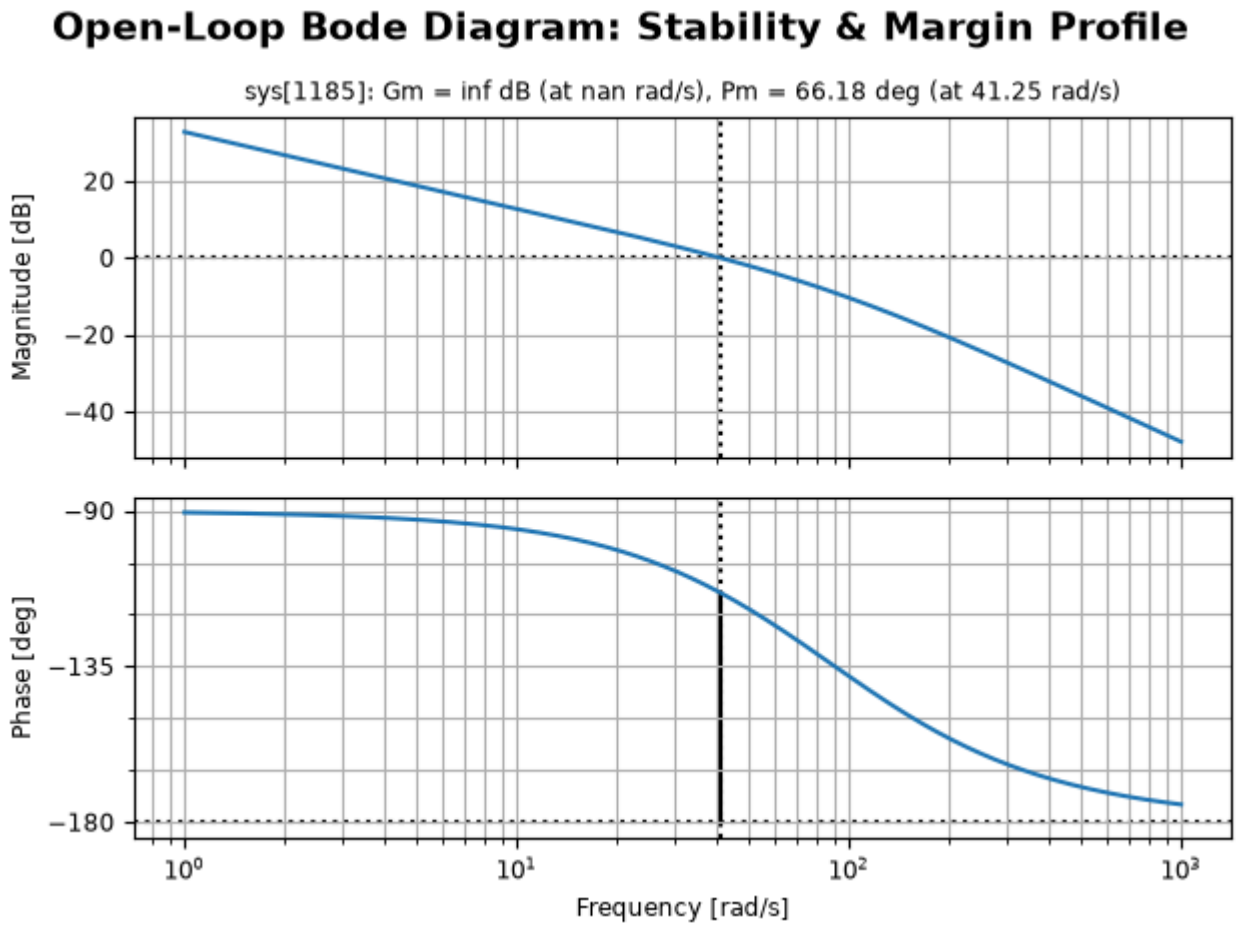


Figure 1-5: Bode plot of compensated Open-loop for roll rate

Time-domain Transient Responses:

From **Figure 1-6** we see that the rise time is just at $50.8ms$ which is within the target rise time of $80ms$. It is expected from type-1 system not to have steady-state error from a step response, nonetheless it is maintained with this implemented lead compensator. On the other hand, the velocity error from ramp input is $0.0286\ \mathrm{rad/s}$ and remains constant.

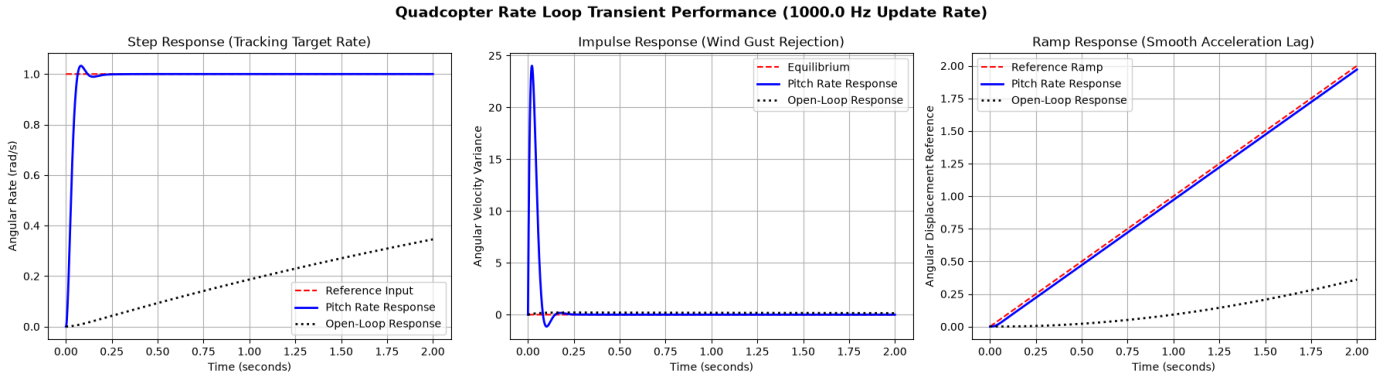


Figure 1-6: Step, impulse, and ramp input responses for pitch-rate closed-loop system

Similarly, from **Figure 1-7**, the rise time is just at $42.0\ ms$ which is within the target rise time. Reference tracking via steady state error performs as required and the velocity error from ramp input is $0.0228\ \mathrm{rad/s}$ and remains constant.

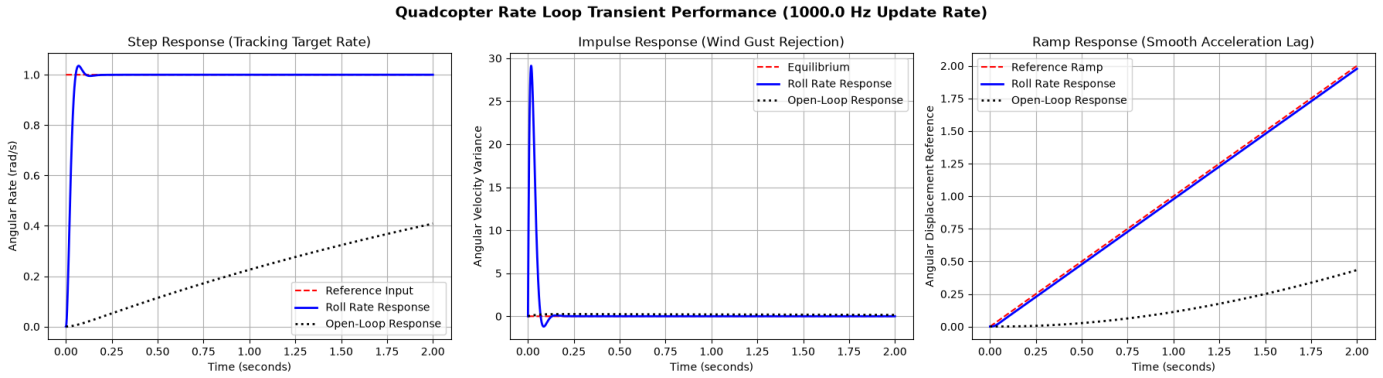


Figure 1-7: Step, impulse, and ramp input responses for roll-rate closed-loop system

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7- Robustness and Uncertainty

Gain and phase margin measure instability along specific axes or angles. The vector margin metric on the other hand gives a comprehensive composite measure of robustness against combined gain and phase variations.

So, designing with a vector margin that aims to keep the loop transfer function $L(j\omega)$ away from the critical point $(-1 + j0)$.

The vector margin is the inverse of the supremum (maximum peak) of the sensitivity transfer function.

$$VM = \min_{\omega} |1 + L(j\omega)| = \frac{1}{\max_{\omega} |S(j\omega)|} = \frac{1}{M_s}$$

Designing for a good vector margin is identical to minimizing the maximum peak of the sensitivity function M_s. A vector margin in the range of $[0.5, 0.7]$ (which corresponds to an M_s of $[1.4, 2.0]$ is a robust range of values.

Robust control methods (e.g. H_∞ sensitivity based loop-shaping) with model-order reduction guaranteens robustness, but these methods are overkill for stability augmentation or angular rate compensation for drones, unless stability guarantees on extreme maneuvering is a requirement.

Another way of designing for robutness is reformulating the problem as an optimal controler (e.g. LQG) or direct PID Tuning via constrained optimization instead of heuristic approaches (e.g. Ziegler-Nichos). The constrained optimization of PID gains would aim in its objective cost function to maximize tracking performance (like minimizing Integrated Absolute Error), subject to a strict non-linear constraint:

$$M_s = \max_{\omega} |S(j\omega)| \leq 1.6$$

With $S(j\omega)$ being the sensitivity function.

This provide specific guarantees that your PID controller achieves the fastest possible response time with the target vector margin in mind.

There is a fundamental tradeoff baked in the following identity condition.

$$S(j\omega) + T(j\omega) = 1$$

Where $T(j\omega)$ is the closed-loop transfer function.

At low frequencies, where good tracking should be achieved, $L(j\omega)$ needs be large. This makes $S(j\omega) \approx 0$ (good disturbance rejection) and $T(j\omega) \approx 1$ (idea setpoint tracking). While, at high frequencies, $L(j\omega)$ needs be small to filter out high-frequency signals. This makes $S(j\omega) \approx 1$ and $T(j\omega) \approx 0$ (good noise filtering).

Another important consideration to design the controller for are frequency specific noise-rejection such motor vibrations. One to way to improve noise/viration rejection in this respect is to employ a notch filter at the particular vibration mode without attenuating significantly the bandwidth.